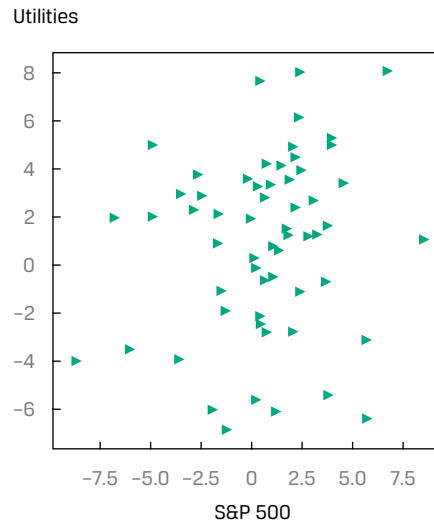


Exhibit 25: Scatter Plot of Utilities Sector Index Return versus S&P 500 Index Return



Despite their relatively straightforward construction, scatter plots convey valuable information. First, it is important to inspect for any potential association between the two variables. The pattern of the scatter plot may indicate no apparent relationship, a linear association, or a non-linear relationship. Furthermore, the strength of the association can be determined by how closely the data points are clustered around a line drawn across the observations.

Examining Exhibit 24 we can see the returns of the IT sector are highly positively associated with S&P 500 Index returns because the data points are tightly clustered along a positively sloped line. In contrast, Exhibit 25 tells a different story for relative performance of the utilities sector and S&P 500 index returns: The data points appear to be distributed in no discernable pattern, indicating no clear relationship among these variables.

Second, observing the data points located toward the ends of each axis, which represent the maximum or minimum values, provides a quick sense of the data range. Third, if a relationship among the variables is apparent, inspecting the scatter plot can help to spot extreme values (i.e., outliers). For example, an outlier data point is readily detected in Exhibit 24 as indicated by the arrow. Finding these extreme values and handling them with appropriate measures is an important part of the financial modeling process.

Scatter plots are a powerful tool for finding patterns between two variables, for assessing data range, and for spotting extreme values. In some situations, however, we need to inspect for pairwise associations among many variables—for example, when conducting feature selection from dozens of variables to build a predictive model.

Now that we have some understanding of sample variance and standard deviation, we can more formally consider the concept of correlation between two random variables that we previously explored visually in the scatter plots.

Covariance and Correlation

Correlation is a measure of the linear relationship between two random variables. The first step in considering how two variables vary together, however, is constructing their covariance.